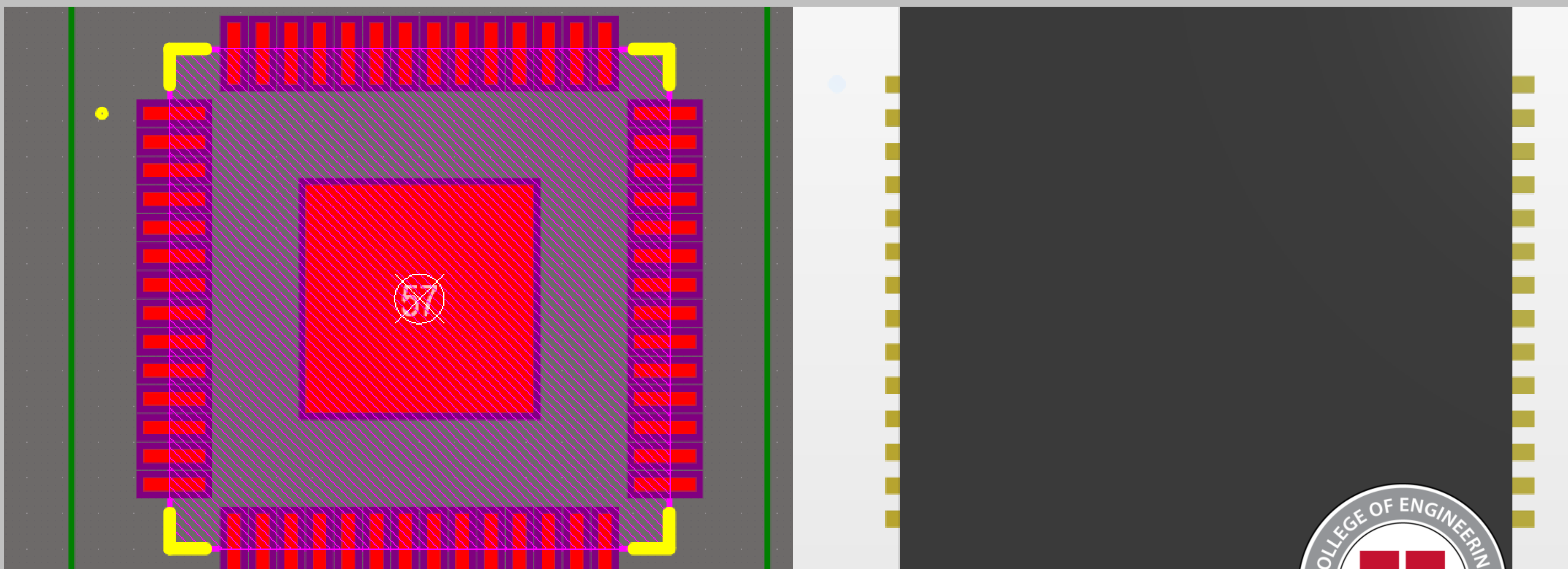
IEEE PCB Design Workshop: (Week 04)

Footprints and PCB Layout



Hosted By: Adrian Sucahyo and IEEE at the University of Utah

Adapted From: IEEE x FSAE Workshop SP25 with Nick Howard and Adrian Sucahyo

Workshop Outline

Schedule:

- Feb. 17 – Introduction to Schematics
- Feb. 24 – Datasheets and Components
- Mar. 3 – Introduction to PCB Layout
- ** SPRING BREAK **
- Mar. 17 – Layout Continued
- Mar. 24 – Soldering Week 1
- Mar. 31 – Soldering Week 2
- Apr. 3 – Soldering Week 3

Announcements

- Alternative Projects
  - We will be able to get alternative project boards manufactured if submitted by the deadline.
  - Limited to the 10 cm x 10 cm dimensions outlined by JLCPCB.
  - Talk or email me if you have any questions!

- PCB Submission Deadlines
  - For both workshop and custom PCBs, the deadline for submission is **March 20th**

Want more experience?

- Consider joining the FSAE tractive team!
  - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
  - No experience required!

U of U FSAE Discord Link

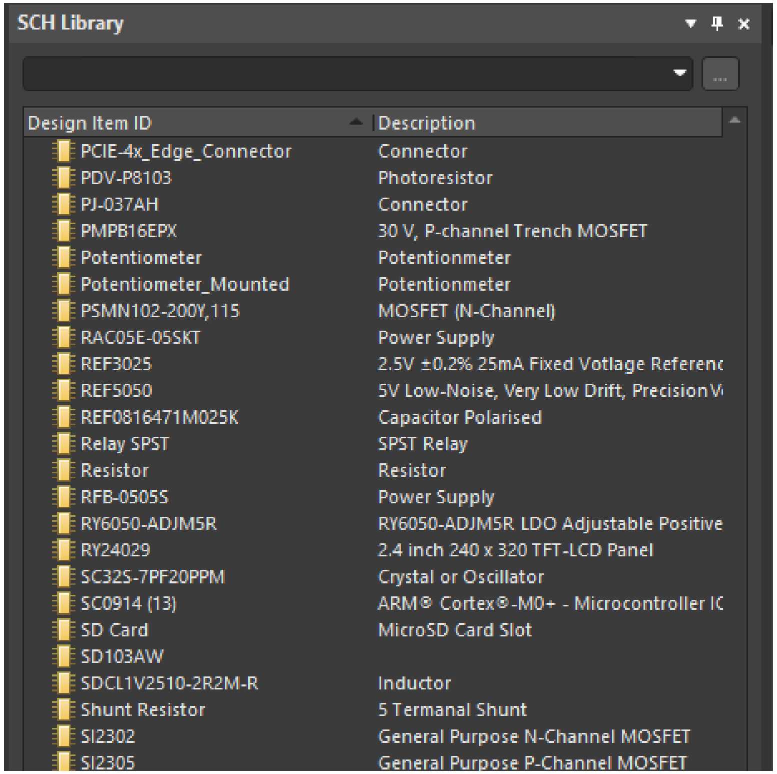
Join the IEEE Discord

- If you haven’t already, please join the IEEE Discord server for additional information and updates regarding this workshop

U of U IEEE Discord Link

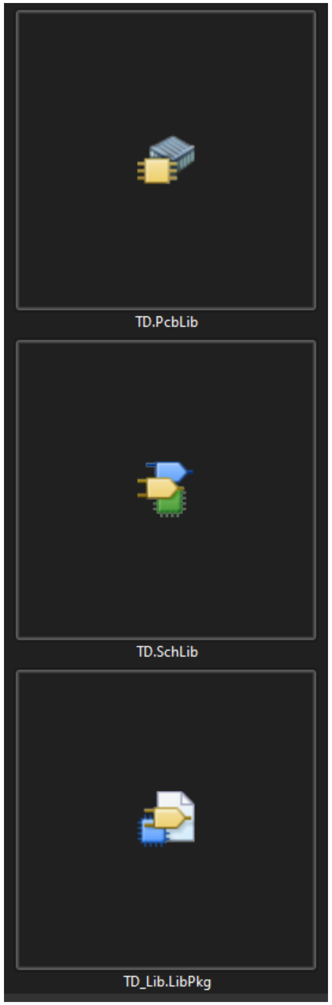
Component Libraries Refresher

- Libraries are collections of components and footprints
  - All components for a project must be derived from a library
  - Projects may reference multiple different component libraries
  - Relatively consistent across platforms
- Typically, there are symbol and footprint libraries
  - Tightly coupled together



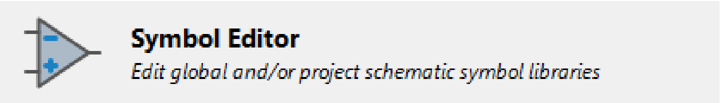
Component Library Types (Altium)

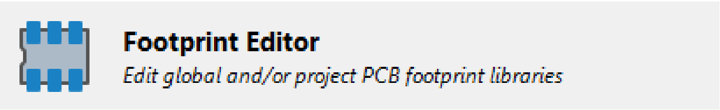
- .SchLib
  - Contains all component schematic symbols
- .PcbLib
  - Contains all footprint information
- .LibPkg
  - Also known as an integrated library
  - Packages multiple .SchLib and .PcbLib libraries together for easier management

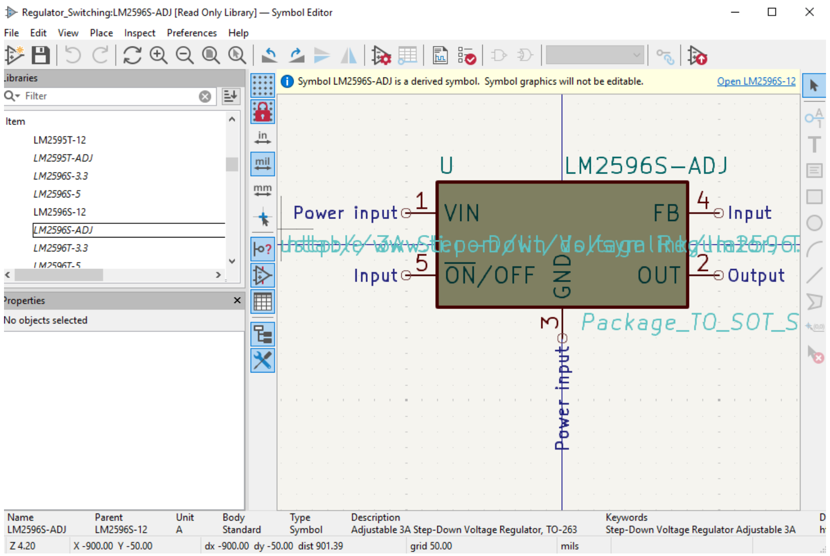


Component Library Types (KiCAD)

- .kicad_sym
  - Contains all the symbol information for components in the library
- .pretty
  - Contains all the footprint information to be used with components







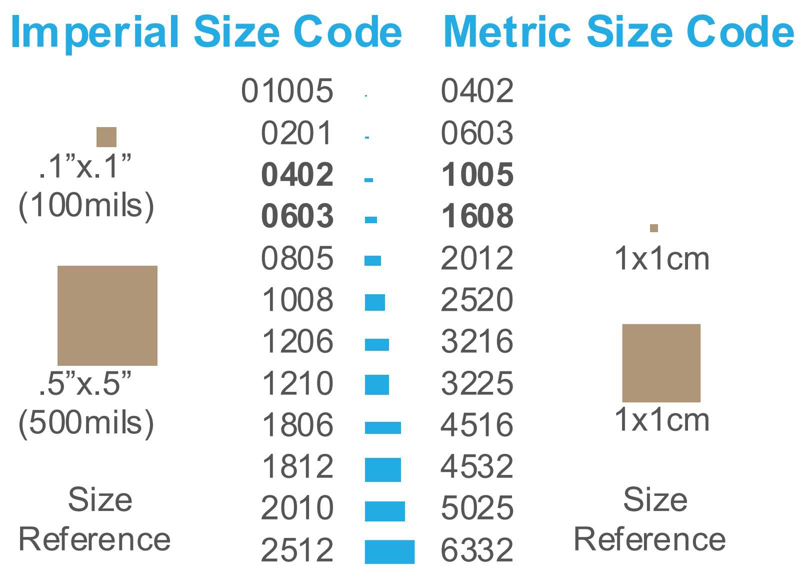
Standard vs. Non-Standard Footprints

- Many components are in "standard" packages
  - Key parameters are standardized across manufacturers
- Certain components are in "non-standard" footprints due to custom features or other restrictions

https://www.indiamart.com/proddetail/smd-electronic-component-2856299588912.html

Resistors and Capacitors

- Many components are in “standard” packages
  - Key parameters are standardized across manufacturers
- Certain components are in “non-standard” footprints due to custom features or other restrictions

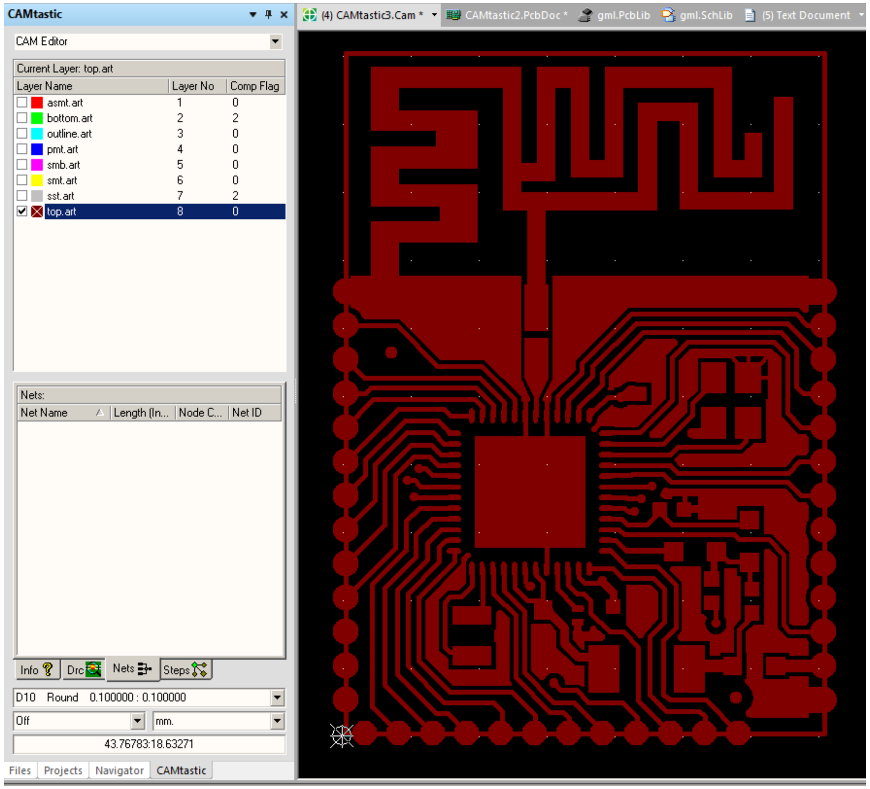


Component Sizes (common in bold)

https://developer.wildernesslabs.co/Hardware/Reference/Components/Packages_and_Sizes/

Gerber Files

- Gerbers (or Gerber Files) are the files sent to the manufacturer that will be used to make the boards
- Engineers at the fab review files for manufacturability and then send it off to production.
- Gerber files can be generated from EDA software but needs to be reviewed.



https://www.onethesis.com/gerbers-to-footprint/

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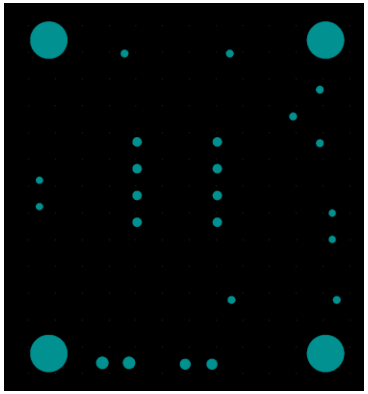
NC Drill Files

- NC Drill Files contain the information for the drill holes of the PCB.
  - These are generated separately from the Gerber Files



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-----------------------------------------------------------------------------------------------------------------
NCDrill File Report For: Workshop Project.PcbDoc   10/29/2024  1:09:38 AM
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Layer Pair : Top Layer to Bottom Layer
ASCII RoundHoles File : Workshop Project.TXT

Tool       Hole Size               Hole Tolerance             Hole Type       Hole Count   Plated         Tool Travel
-----------------------------------------------------------------------------------------------------------------
T1      28mil  (0.7mm)                                          Round            4          PTH        1.29inch (32.82mm)
T2      30mil  (0.75mm)                                         Round            7          PTH        2.02inch (51.35mm)
T3      35mil  (0.9mm)                                          Round            8          PTH        0.90inch (22.86mm)
T4      41mil  (1.05mm)                                         Round            2          PTH        0.10inch (2.54mm)
T5      47mil  (1.19mm)                                         Round            2          PTH        0.10inch (2.54mm)
T6      140mil  (3.556mm)                                       Round            4          PTH        3.24inch (82.30mm)
-----------------------------------------------------------------------------------------------------------------
Totals                                                                          27

Total Processing Time (hh:mm:ss) : 00:00:00
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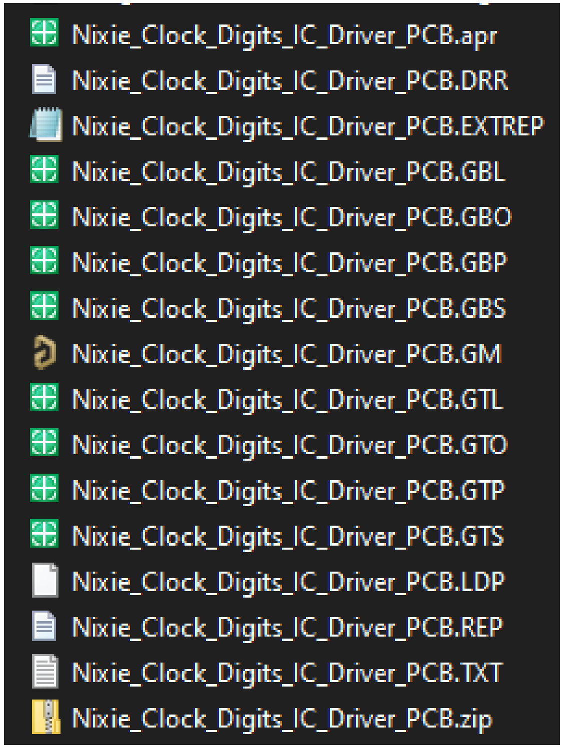
Gerber File and Drill File Types

- Each file represents a different layer or operation type.
- Common files:
  - .gto, .gbo = top/bottom silkscreen
  - .gts, .gbs = top/bottom solder mask
  - .gtp, .gbp = top/bottom solder paste
  - .gtl, .gbl = top/bottom copper
  - .gm = board outline
  - .drl / .txt = CNC drill



Questions?

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Download Today's Project Files

Navigate to the workshop GitHub and download today's files

https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Spring-2026